CENG 252, Fall 2026, HW#2

Instructor: Zheng Chen (Email: zhc199@ucsd.edu)

Due 8:59pm, 10/17/2026

Problem 1: The elementary gas-phase reaction

$$(CH_3)_3COOC(CH_3)_3 \rightarrow C_2H_6 + 2CH_3COCH_3$$

is carried out isothermally at 400K in a flow reactor. The specific reaction rate at 50°C is 10^{-4} min^{-1} (from pericosity data) and the activation energy is 85 kJ/mol. Pure di-tert-butyl peroxide enters the reactor at 10 atm and 127°C and a molar flow rate of 2.5 mol/min. Plot X, y as a function of the PFR volume when $\alpha = 0.001$ dm^{-3}. What are the X, and y at V = 500 dm^3 ?

Problem 2: The irreversible elementary gas-phase reaction

$$A + B \rightarrow C + D$$

is carried out isothermally at 305K in a packed-bed reactor with 100kg of catalyst. The entering pressure was 20 atm and the exit pressure is 2 atm. The feed is equal molar in A and B and the flow is in the turbulent flow regime, with $F_{A0} = 10\ mol/min$ and $C_{A0} = 0.4\ mol/L$. Currently 80% conversion is achieved.

(1) What would be the conversion if the catalyst particle size was doubled and everything else remained the same?

(2) What would be the conversion if the exit pressure is 4 atm and everything else remained the same?

Problem 3: An acid-catalyzed irreversible liquid-phase reaction A → B is carried out adiabatically in a CSTR. The reaction is second order in A. The feed is equimolar A and solvent (S, which contains catalyst), and enters the reactor at a total volumetric flow rate of 10 dm^3/min at a concentration of A of 4 mol/L. The feed enters at 300 K. The product and reactant heat capacities are 15 cal/mol °C, the solvent is 18 cal/mol °C. The reaction rate constant at 300 K is 0.0005 dm^3/mol min. The reaction has an activation energy of 15,000 cal/mol and a heat of reaction of -3300 cal/mol °C.

(a) What V_{CSTR} is needed to achieve 80% conversion, and at what temperature does it operate?

(b) What conversion can be achieved in a 10^3 dm^3 CSTR, and what is the new exit temperatur